

NOISE POLLUTION AND ITS EFFECTS ON HUMAN MENTAL AND PHYSICAL HEALTH

An Undergraduate Thesis

By

Seth Anderson

Presented to the Faculty of

The Environmental Studies Program at the University of Nebraska-Lincoln

In Partial Fulfillment of Requirements

For the Degree of Bachelor of Science/Arts

Major: Environmental Studies

Minor: Psychology

Under the Supervision of

Thesis Advisor: Dr. Heather Akin

Thesis Reader: Larry Carlson

Lincoln, Nebraska

May 2022

ABSTRACT

Noise pollution is defined as unwanted sound, capable of causing both mental and physical harm. In order to see if there was a correlation between increasing noise pollution (dB) and (a) increasing Negative Affect Score (NAS), (b) decreasing Positive Affect Score (PAS), and (c) increasing Resting Heart Rate (RHR), research was conducted. Participants were taken to four different testing locations, each with a corresponding dB level, and asked to (a) have their Resting Heart Rate (RHR) taken and (b) fill out a questionnaire based on the Positive and Negative Affect Schedule (PANAS) psychological measure. At the conclusion of this study, it was shown that (a) there was a significant effect of noise pollution (dB) on Negative Affect Score (NAS), (b) there was NOT a significant effect of noise pollution (dB) on Positive Affect Score (PAS), and (c) there was a significant effect of noise pollution (dB) on Resting Heart Rate (RHR). Since this study was limited by different factors (weather conditions, sample size, varying moods, and varying RHR), different steps could be taken to tighten the research design. Overall, due to the small sample size (10), it is not statistically significant enough to draw any concrete conclusions.

Keywords: noise pollution, mental health, physical health, resting heart rate, PANAS, positive affect score, negative affect score

ACKNOWLEDGEMENTS

First, I would like to thank my thesis advisor Dr. Heather Akin for her continued support over the course of this project; her extensive knowledge and past research experience helped facilitate any question I had. Second, I would like to thank my thesis reader Larry Carlson for proofreading my work and providing valuable insight along the way. Third, I would like to thank all the participants for taking part in this research. Your participation and time commitment were much appreciated. And last of all, I would like to thank Dr. Dave Gosselin for his patience and for providing me with guidance over the course of this project; it would not have been impossible without all your help.

TABLE OF CONTENTS

Abstract.....	2
Acknowledgments.....	3
Introduction.....	6
Objective and Hypotheses.....	12
Materials and Methods.....	13
Results.....	17
Discussion.....	21
Limitations.....	22
Conclusion.....	23
Recommendations.....	24
References.....	26
Appendix A: Survey Questions.....	29
Appendix B: Raw Participant dB and RHR Data.....	31
Appendix C: Raw Participant PANAS Data.....	32

TABLE OF TABLES

Table 1: Common sources of noise and respective dB levels.....	7
Table 2: NIOSH dB standards and respective 100% noise dose exposure time.....	8

TABLE OF FIGURES

Figure 1: Map showing noise pollution levels over a 24-hour period in Lincoln, Nebraska.....	11
Figure 2: Map showing the four different testing locations in Lincoln, NE.....	14
Figure 3: Line graph showing the relationship between each testing location and Positive and Negative Affect Schedule (PANAS) scoring.....	18
Figure 4: Line graph showing the relationship between each testing location and Resting Heart Rate (RHR).....	20

INTRODUCTION

What is noise pollution? This is the first question that must be addressed. “Noise is defined as unwanted sound.” There are countless sources of noise. Mechanical appliances, entertainment systems, speakers, cell phones, transport vehicles like cars, airplanes, trains, and boats, mechanical equipment like mowers, air-conditioners, drills, and large equipment used for construction are examples of different sources of noise (Gupta et. Al, 2018). Pollution, on the other hand, can be defined as the introduction of contaminants into the environment capable of causing negative effects and harm. Combining the two, noise pollution can be defined as all unwanted noise that occurs in a setting capable of causing harm (Goines & Hagler, 2007). With an ever-expanding industrial world, the prevalence of noise pollution will only continue to grow. Noise pollution exposure has been attributed to different mental health effects like sleep disturbances, anxiety, concentration issues, and physical health effects like hearing impairment and cardiovascular disturbances.

Noise is described both in terms of intensity (loudness) and frequency (pitch). Both of these metrics are used in calculating potential damage noise can have on someone’s hearing. “Sound intensity is measured as sound pressure level (SPL) in a logarithmic decibel (dB) scale” (Rabinowitz, 2000). The dB scale is often weighted toward the higher end of frequencies, noises that are more capable of causing damage like gunshots and airplanes. Airplanes taking off hover around 140 dB, gunshots hover anywhere from 140-170 dB, and rock concerts can reach 110-120 dB. 60 dB is common for normal conversations, and whispering is roughly 30-40 dB. Permanent hearing loss is common in frequent exposures of 85 dB+ for an eight-hour period (Morata et. al, 1993).

Common Sources of Noise	
SOUND	LOUDNESS (DB)
Gunshot (peak level)	140 to 170
Jet takeoff	140
Rock concert, chain saw	110 to 120
Diesel locomotive, stereo headphones	100
Motorcycle, lawnmower	90
OSHA level for hearing conservation program	85* (8-hour time-weighted average)
Conversation	60
Quiet room	50
Whisper	30 to 40
OSHA = Occupational Safety and Health Administration; dB = decibels.	

Table 1: Common sources of noise with respective dB levels.

Note. Rabinowitz, P. (2000). Noise-induced hearing loss. American family physician, 61(9), 2749-2756.

Time to reach 100% noise dose	Exposure level per NIOSH REL
8 hours	85 dB(A)
4 hours	88 dB(A)
2 hours	91 dB(A)
60 minutes	94 dB(A)
30 minutes	97 dB(A)
15 minutes	100 dB(A)

Table 2: NIOSH dB standards and respective 100% noise dose exposure time.

Note. CDC. (2018, February 6). *Reducing noise exposure: Guidance & Regulations*. Centers for Disease Control and Prevention.

Gathered from the CDC website, this table shows different dB levels and corresponding time dosages before permanent hearing damage begins to occur. Specifically, this table deals with noise level exposure in workplace environments. While this table deals with noise pollution levels in workplace environments, it is a good metric in order to understand how dB levels have an effect on our hearing. In order to protect workers against the adverse effects of noise pollution, the United States passed different occupational regulations to mitigate these issues.

NIOSH established a baseline value for this table at 85 dB. At 85 dB, workers are allotted a maximum of 8 hours of noise exposure. Beyond this level, permanent hearing damage can occur, and for every 3 dB increase, this exposure time is cut in half (CDC, 2018). As shown in

Table 1, diesel locomotives have dB levels hovering around 100. This dB level has the potential to cause permanent hearing damage in a short amount of time (>15 minutes). And with jet takeoffs having dB levels close to 140, this noise level can cause permanent hearing damage immediately. In fact, 110 dB is considered the “threshold of discomfort,” whereas 130 dB is considered the “threshold of pain” (Informed Health, 2008). Anything beyond 130 dB can cause immediate hearing damage.

There have been numerous physical health effects seen in people exposed to noise pollution. Noise pollution has been connected to different heart conditions in humans. People living in noisy environments get used to the constant noise, but our hearts are not as capable. The sympathetic nervous system is activated while sleeping, constantly changing us from deep and light sleep. In fact, noise activates the body’s fight-or-flight mechanism, increasing blood pressure (Hammer et. al, 2014). Constant activation of the nervous system puts strain on our cardiovascular system, disrupting sleep quality and increasing risk of developing heart conditions (Sforza et. al, 2004). Specifically, in a study conducted in a major industrial city in Pakistan, respondents reported high levels of headaches, hypertension, and high blood pressure (Farooqi et. al, 2020).

Cardiovascular strain is not the only physical health effect associated with exposure to noise pollution. Chronic exposure to noise pollution can have serious effects on hearing. Noise-induced hearing loss (NIHL) is of the more serious types (Hammer et. al, 2014). “Noise-induced hearing loss is the second most common sensorineural hearing loss, after age-related hearing loss (presbycusis)” (Rabinowitz, 2000). Roughly 5% of the global population suffers from NIHL, and around 50% of those cases are attributed to noise exposure. One of the first onsets of NIHL is tinnitus. Tinnitus can be characterized by abnormal ringing like whistling, grinding, hissing, or

buzzing that occurs in the body. Tinnitus can have effects on daily activities like sleeping, studying, and working (Gupta et. al, 2018).

As well as the physical effects seen, there are numerous psychological effects associated with extended exposure to noise pollution. One of the most intrusive effects is its influence on psychomotor effects like stress, sleep disturbances, concentration deficiencies, irritability, violent tendencies, and other psychological effects. In one such study conducted in Kuwait, residents in neighborhoods exposed to traffic noise had significantly higher sensitivity to noise levels, elevating mental health disturbances and affecting overall sense of well being (Al-Mutairi et. al, 2011). In another study conducted in Iran, nervousness, sleep quality disturbances, hypertension, and depression were among some of the mental health effects caused by noise pollution (Geravandi et. al, 2015).

Some of the more notable effects have been seen in children and their learning environments. Children are among a group that's more vulnerable to non-physical health effects. Considering children are still developing physically and emotionally, noise pollution in their personal environment can have different negative effects associated with it. One of the most common effects seen in children is cognitive function hindrance. This affects their central processing and language comprehension, essential skills in a learning environment (Haines et. al, 1998). One controlled study compared children in two different noise environments: one exposed to high levels of railway noise and one exposed to minimal noise, a quiet classroom. At the conclusion of the study, children in the noisy environment scored significantly less on tests and were calculated to be almost 4 months behind the children in a quiet classroom (Bronzaft and McCarthy, 1975).

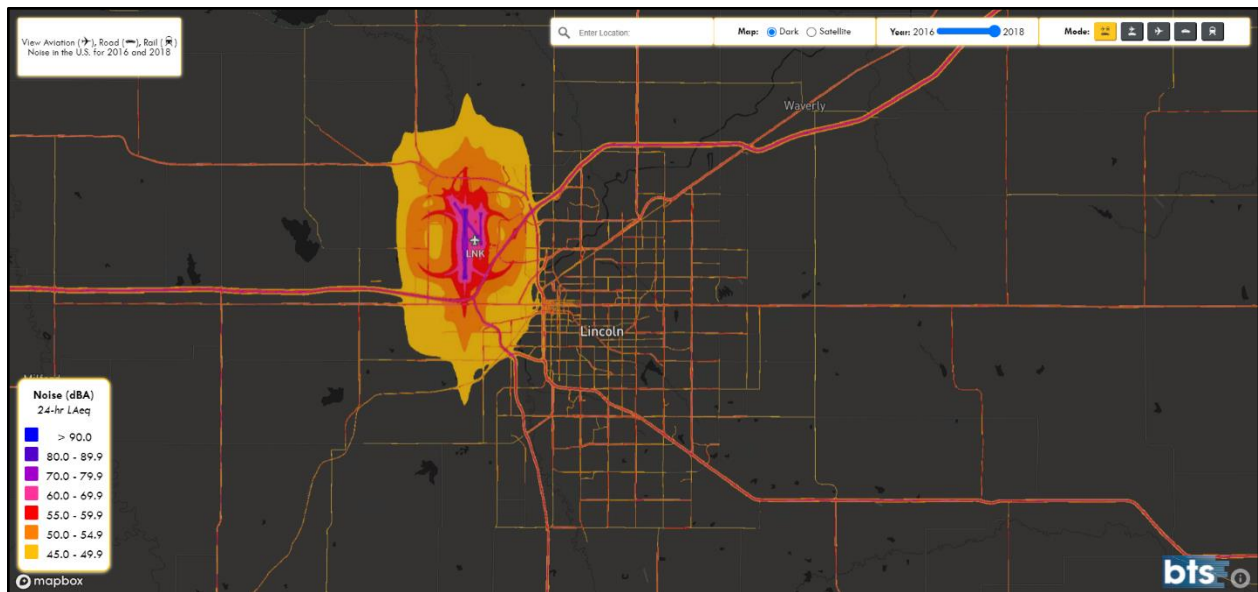


Figure 1: Map showing noise pollution levels over a 24-hour period in Lincoln, Nebraska.

Note. Bureau of Transportation. (2018). National Transportation Noise Map. Retrieved March 23, 2022.

Collected from the Bureau of Transportation’s website, this figure shows average dB levels over a 24-hour period in Lincoln, NE, the testing location for this research. It is not used to evaluate noise levels at certain times or in certain locations, but it is useful in examining noise trends. Additionally, since it is gathered from the Bureau of Transportation, it focuses on transportation and does not include other forms of noise pollution. As seen in the image, most of the extreme noise pollution comes from the Lincoln airport (LNK) and surrounding area, reaching dB levels of >90.0. Much of the other noise pollution is condensed around major highways and streets within Lincoln. Due to the University of Nebraska-Lincoln’s close proximity to major highways and the Lincoln airport (LNK), it is especially prone to the effects of noise pollution. Further information on this noise pollution tool and how to properly analyze it can be found here: <https://www.bts.gov/geospatial/national-transportation-noise-map>

Academic research in this area is of utmost importance. As many as 30 million Americans are exposed to noise pollution at a harmful level (Rabinowitz, 2000). Physical health effects like cardiovascular disease and permanent ear damage are concerning, but psychological effects like decreases in cognitive function, sleep disturbances, stress, and anxiety are equally as concerning. Even more outside of the United States are exposed to constant levels of noise pollution. With an ever-expanding technological and industrial world, the effects of noise pollution will only continue to grow. While there has been quite a bit of research conducted in this field already, the need for more research will only continue to grow as global population increases and noise levels rise.

Objective and Hypotheses

This study seeks to find a correlation between noise pollution and how it affects human mental and physical health, which has led to the research questions: how does noise pollution (dB) affect human emotions and mental health? Is there a correlation between increased noise pollution levels (dB) and increased mental disturbances? How does noise pollution (dB) affect humans physically? Is there a correlation between increased noise pollution (dB) and the presence of physical health effects like increased heart rate?

H1: As noise pollution (dB) increases, (a) Negative Affect Score (NAS) will increase and (b) Positive Affect Score (PAS) will decrease.

H2: As noise pollution (dB) increases, physical effects like Resting Heart Rate (RHR) will increase.

MATERIALS AND METHODS

This research took place in the spring semester of 2022, mainly taking place over a period of four weeks and seeks to understand the correlation between noise pollution (dB) levels in the environment and how that affects human mental and physical health. In order to effectively test the correlation between increasing noise pollution (dB) levels and its effects on human mental and physical health, research was conducted. This research captured participants' Resting Heart Rate (RHR), self-reported Negative Affect Score (NAS), and self-reported Positive Affect Score (PAS) at each testing location, with each location corresponding with a measured noise pollution (dB) level.

Different factors were considered when choosing the appropriate testing conditions. Some of the considerations were (a) time of day, e.g., peak traffic hours (06:00 to 19:00) vs. nonpeak traffic hours; (b) time of week, e.g., weekday vs weekend; and (c) relative dB noise level e.g., whether conditions at each location differed in noise level enough to show correlation between increasing dB and emotional disturbance and increased heart rate. Given these factors, the tests were performed from 12:00 to 18:00 on weekdays, Monday to Friday. Peak traffic hours can vary by country, city, and season, but in order to capture the largest range of peak traffic, this range was chosen for the research. Additionally, relative dB noise levels can vary even within the same timeframe, but this fluctuation is just a representation of natural traffic conditions.

The experiment was conducted at four different testing locations within Lincoln, Nebraska. Testing locations were chosen prior to testing based on perceived loudness and noise pollution. Preliminary noise pollution (dB) levels were gathered prior to confirm pre existing noise perceptions but were not used in the final data. Final testing locations are as follows:

- **Location #1:** Located inside Love library on University of Nebraska-Lincoln city campus, a setting with quiet ambient noise. Average aggregate dB: 44.8.
- **Location #2:** Located outside of University of Nebraska-Lincoln city campus union in the park, a setting with faint noise pollution. Average aggregate dB: 62.7.
- **Location #3:** Located along 10th Street on the west side of University of Nebraska-Lincoln city campus outside of the architecture building, a busy road with noisy, frequent traffic. Average aggregate dB: 68.4.
- **Location #4:** Located along N 21st and O Street, a road with very high traffic and high levels of noise pollution. Average aggregate dB: 84.2.

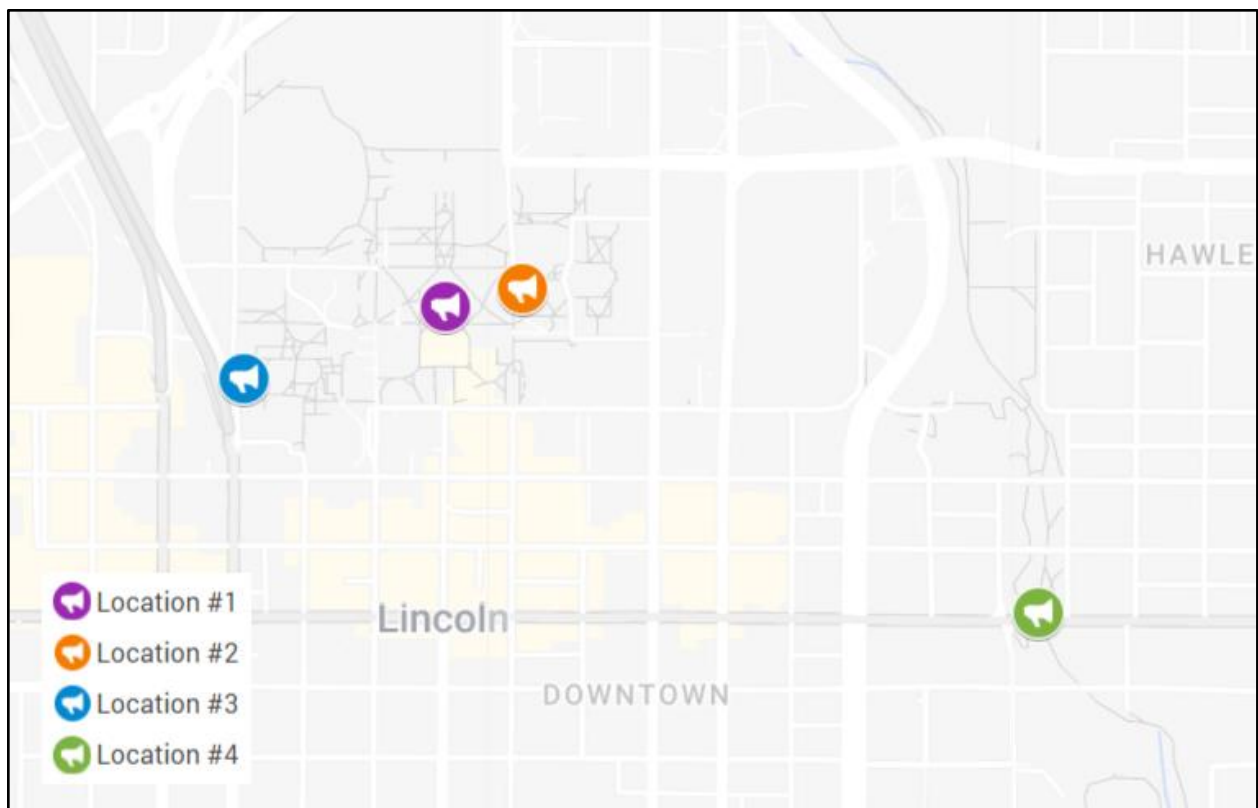


Figure 2: Map showing the four different testing locations in Lincoln, NE.

At each location, an initial dB level was gathered. This was measured for 5 minutes, gathering the following values: maximum dB, minimum dB, and average dB. This data was averaged among all participants at each location and presented in aggregate form in the final data. However, maximum dB and minimum dB were not used to support any conclusions, but rather used as a reference value for each location. After acquiring the initial dB values, participants were then asked to wear a heart rate monitor. With the heart rate monitor equipped, participants' heart rates were measured for 5 minutes; values were recorded at each minute interval and averaged, leaving us with an average Resting Heart Rate (RHR) for each participant at each location. While heart rate was being monitored, participants were also asked to fill out a questionnaire based on the Positive and Negative Affect Schedule (PANAS). As stated above, this was repeated at each location, and once all the data was collected, the trial was concluded. In the end, we were left with the following values: maximum dB, minimum dB, average dB, average Resting Heart Rate (RHR), and Positive and Negative Affect Schedule (PANAS) questionnaire scoring.

The survey instrument in this research makes use of a questionnaire acquired from PsyToolKit (<https://www.psytoolkit.org/>) and formatted into Google forms. The questionnaire was based on the Positive and Negative Affect Schedule (PANAS), a scale measuring different feelings and emotions at a given moment (Riopel, 2022). *Positive* corresponds with positive emotions like joy and cheerfulness, whereas *negative* corresponds with negative emotions like fear and sadness. The questionnaire consists of 20 questions, each consisting of a single word: interested, distressed, excited, upset, strong, guilty, scared, hostile, enthusiastic, proud, irritable, alert, ashamed, inspired, nervous, determined, attentive, jittery, active, and afraid. The 5-point Likert scale is used for scoring and adding up the given values. The range is from 'Very Slightly

to Not at All’ to ‘Extremely’. For each word, participants were asked to select one of the five values. PANAS shows high internal validity and is consistent no matter what time interval is used, however, reliability tends to be higher as the time frame lengthens (Riopel, 2022). Widely used in clinical and community trials, PANAS has been found to be a reliable gauge of positive and negative affect.

Noise levels (dB) were measured using the *Decibel X: dB Sound Level Meter* smartphone application. The application is pre calibrated and in accordance with NIOSH and OSHA standards. Resting Heart Rate (RHR) levels were measured using a Coospo chest heart rate monitor. Participants were asked to equip the heart rate monitor themselves. In order to effectively use the *Coospo* heart rate monitor, another application was used: *Strava*. This application was used to display live heart rate feedback.

Participants taking part in this study have been made confidential, and there were numerous steps taken to ensure that confidentiality. There was minimal risk involved in participating, and the study was conducted entirely on a voluntary basis. While data being collected is anonymous to the public, they are identifiable to myself and the two people that helped me conduct the research in the form of an ID number. Furthermore, the raw data in the appendix section has been “de-identified” and anonymized further as another step to protect participants’ identities. In instances where multiple participants were being tested at the same time, data was kept private from other participants and was only accessible to each individual. Additionally, access to the data is limited only to myself and the two people that helped me with the research.

RESULTS

This research seeks to understand the impact of noise pollution, specifically the effects noise pollution has on human mental and physical health in the forms of positive and negative emotions and resting heart rate. At the conclusion of this study, there were 10 participants in total; half of the participants were male (5), and the other half of the participants were female (5). The average age of all the participants was 23.

The type of statistical test used with the participant data was within-subjects analysis of variance (ANOVA) in order to test each hypothesis. The hypotheses are as follows:

H1: As noise pollution (dB) increases, (a) Negative Affect Score (NAS) will increase and (b) Positive Affect Score (PAS) will decrease.

- **H1(a):** The results for Negative Affect Score (NAS) - $F(1,9) = 12.75$, $p < 0.01$, showing there is a significant difference between Negative Affect Score (NAS) as noise pollution (dB) levels increase within participants.
- **H1(b):** The results for Positive Affect Score (PAS) - $F(1,9) = 2.76$, $p < 0.13$, showing there is not a significant difference between Positive Affect Score (PAS) as noise pollution (dB) levels increase within participants.

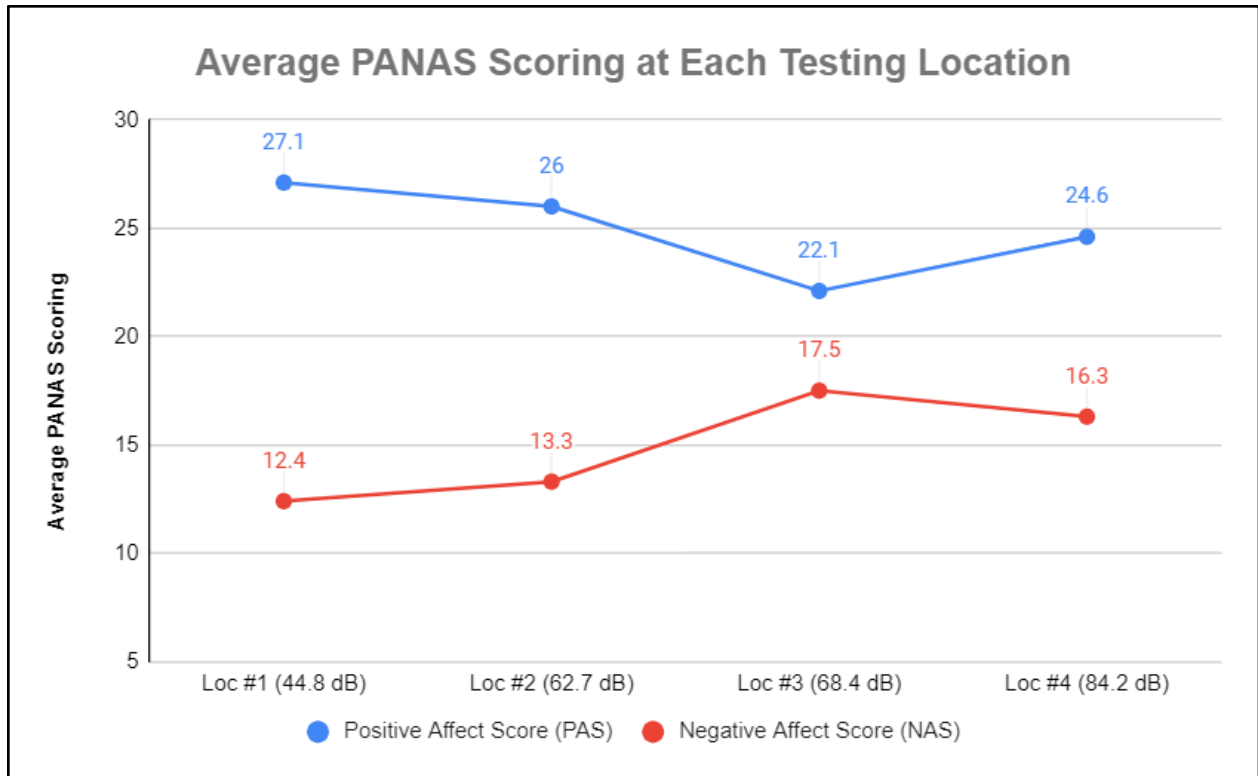


Figure 3: Line graph showing the relationship between each testing location and Positive and Negative Affect Schedule (PANAS) scoring.

At location #1, the average Positive Affect Score (PAS) among all the participants was 27.1. At location #2, the average Positive Affect Score (PAS) among all the participants was 26.0. At location #3, the average Positive Affect Score (PAS) among all the participants was 22.1. And at the final location, location #4, the average Positive Affect Score (PAS) among all the participants was 24.6. In general, there was a negative correlation between average dB level and average Positive Affect Score (PAS). The largest change in average Positive Affect Score (PAS) was between location #1 and location #3.

At location #1, the average Negative Affect Score (NAS) among all the participants was 12.4. At location #2, the average Negative Affect Score (NAS) among all the participants was

13.3. At location #3, the average Negative Affect Score (NAS) among all the participants was 17.5. And at the final location, location #4, the average Negative Affect Score (NAS) among all the participants was 16.3. In general, there was a positive correlation between average dB level and average Negative Affect Score (NAS). The largest change in average Negative Affect Score (NAS) was between location #1 and location #3.

H2: As noise pollution (dB) increases, physical effects like Resting Heart Rate (RHR) will increase.

- **H2(a):** The results for Resting Heart Rate (RHR) - $F(1,9) = 15.32$, $p < 0.01$, showing there is a significant difference between Resting Heart Rate (RHR) as noise pollution (dB) levels increase within participants.

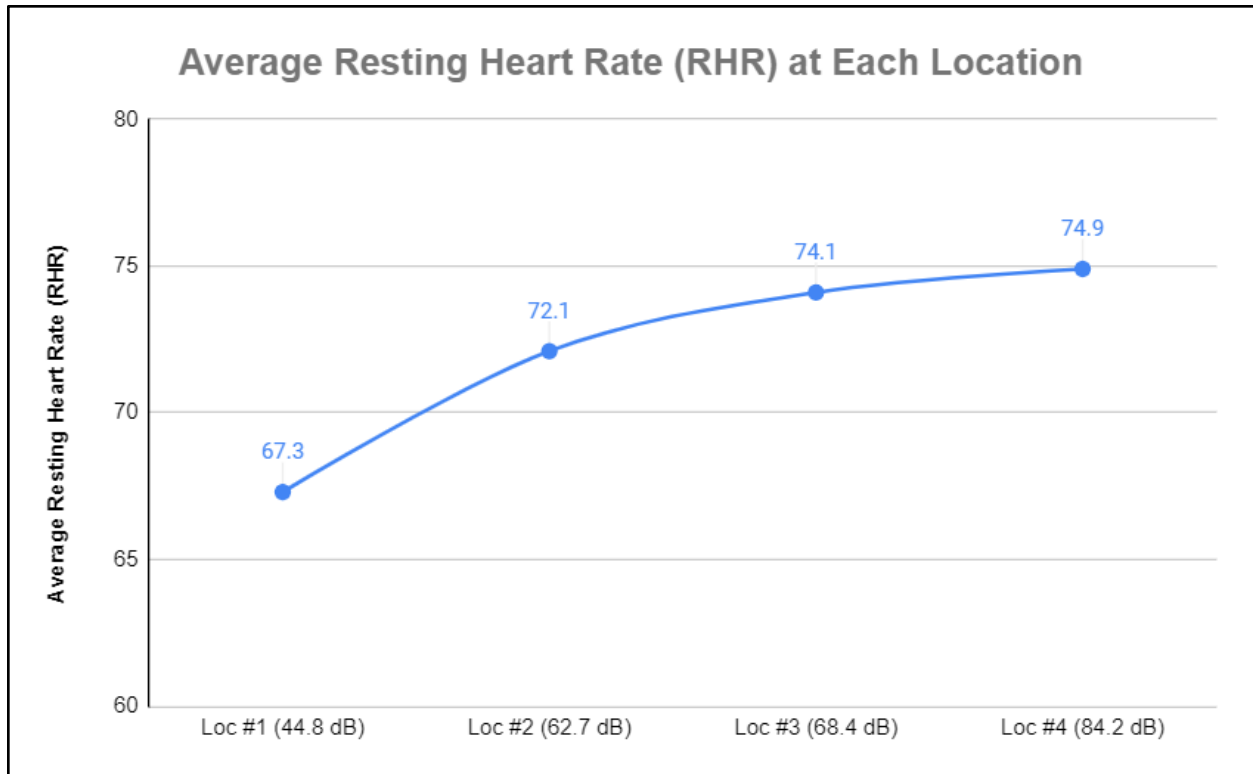


Figure 4: Line graph showing the relationship between each testing location and Resting Heart Rate (RHR).

At location #1, the average Resting Heart Rate (RHR) among all the participants was 67.3. At location #2, the average Resting Heart Rate (RHR) among all the participants was 72.1. At location #3, the average Resting Heart Rate (RHR) among all the participants was 74.1. And at the final location, location #4, the average Resting Heart Rate (RHR) among all the participants was 74.9. In general, there was a positive correlation between average dB level and average Resting Heart Rate (RHR). The largest change in average Resting Heart Rate (RHR) was between location #1 and location #4. The correlation between average Resting Heart Rate (RHR) and average dB corresponds with a Pearson's r value = 0.50, showing positive correlation between the two values (however, not particularly showing strong correlation).

DISCUSSION

While there was statistical significance between increasing noise pollution (dB) levels and (a) increasing Negative Affect Score (NAS) and (b) increasing Resting Heart Rate (RHR), the relatively small sample size makes it hard to draw any concrete conclusions out of it. That being said, it is still important to interpret trends in order to more effectively conduct research in this area. With all the previous research in this field, much of the evidence coincides with the results of this research. In general, there is an increase in mental disturbances and physical health effects like heart rate. While increasing noise pollution (dB) levels and increasing Positive Affect Score (PAS) did not show any statistical significance, it is still important to analyze the results.

In terms of Positive and Negative Affect Schedule (PANAS) scoring, one of the largest variations in the data was between location #1 (44.8 avg dB) and location #3 (68.4 avg dB), and this is especially apparent when looking at Figure 3. Even when taking the noise pollution (dB) differences between location #1 (44.8 avg dB) and location #4 (84.2 avg dB) into account, this was still the largest variance. While there was a statistical significance between increasing noise pollution (dB) levels and increasing Negative Affect Score (NAS), the same was not true for the statistical significance between increasing noise pollution (dB) levels and decreasing Positive Affect Score (PAS). This is an odd discrepancy in the trend. In general, as noise pollution (dB) levels increased, (a) Positive Affect Score (PAS) decreased and (b) Negative Affect Score (NAS) increased, however, both values “peaked” at location #3 (68.4 avg dB). Despite the largest difference between noise pollution (dB) levels being between location #1 (44.8 avg dB) and location #4 (84.2 avg dB), both (a) Negative Affect Score (NAS) decreased and (b) Positive Affect Score (PAS) increased with the 4th trial. Not enough evidence points in any direction as to why this variation exists.

The overall Positive and Negative Affect Schedule (PANAS) scoring was quite varied as well. Some participants' responses were quite varied while others tended to answer without much variability. This could be due to initial mood variance among the participants or due to some flaw in the presentation of the questionnaire. Considering the time of testing coincided with the last couple months of the semester and many of the participants were currently students, stressful feelings and emotions were likely already present.

Limitations

Besides the initial testing location, location #1, every test was conducted outside. When research is conducted in an outdoor environment, the study is subject to different weather effects: ambient temperature, wind, rain, and humidity. All of these weather-related conditions can play a factor in how participants respond.

Since the sample size in this research was small (10), the findings in this research are by no means conclusive. Ideally, the total participant goal in this study was 20-30, but due to time constraints, this goal was not achieved. Studies in this area would benefit from much larger sample sizes in order to draw more concrete conclusions from the data.

As well as that, there are variations by default between different participants. One important variation is in Resting Heart Rate (RHR). Since a normal Resting Heart Rate (RHR) can vary anywhere between 60-100, this can skew the data if someone has a higher-than-average Resting Heart Rate (RHR). As well as that, participants could already be highly accustomed to noise pollution depending on the environment they live in or grew up in. Additionally, since the Positive and Negative Affect Schedule (PANAS) is reliant on current feelings and emotions, participants' current mood could alter the results.

In conclusion, there were implicative trends present in this study, agreeing with much of the past research in this field. But due to numerous limitations and a relatively small sample size (10), some work could be done to improve this research design, specifically in terms of controlling for external factors: weather, temperature, and visual perceptions.

CONCLUSION

This study was conducted in order to find a correlation between increasing noise pollution (dB) and (a) Positive Affect Score (PAS), (b) Negative Affect Score (NAS), and (c) Resting Heart Rate (RHR). Research was conducted in Lincoln, Nebraska at 4 different testing locations, each with a corresponding dB level. At each testing location, participants had their Resting Heart Rate (RHR) taken and were asked to fill out a questionnaire based on the Positive and Negative Affect Schedule (PANAS).

In general, the findings in this research suggest that noise pollution (dB) does have an effect on different emotions, mental well being, and physical feelings. There was a significant effect between increasing noise pollution (dB) levels and (a) increasing Negative Affect Score (NAS) and (b) increasing Resting Heart Rate (RHR). However, due to certain limitations present in this study and the relatively small sample size (10), it is hard to draw any concrete conclusions and lean too heavily in one direction. With a fair amount of research already conducted in this area, mental disturbances and physical health effects will continue to be a concern moving forward. As industry and population continue to grow not only in the United States, but the entire world, research in this area will be of utmost importance and pivotal in protecting humans from the worst effects of noise pollution.

Recommendations

Since this study largely took place outdoors (except location #1), I would recommend conducting research in a controlled environment to see if the effects are the same. In a controlled environment, noise pollution (dB) levels could be controlled instead of relying on external noise pollution (dB) at different locations, which is rather inconsistent. With this research design, one could conduct all testing in a single location free from external noise. Additionally, one would not have to take weather conditions like wind, ambient temperature, and different weather effects into consideration. Since the questionnaire is based on PANAS and relies on participants' current feelings and emotions, weather conditions can play a big factor in how they respond.

While there was no structure to the testing order, the testing did follow a logical order. In general, participants were taken from location #1 to location #4 in increasing order. As discussed in the methods section, noise pollution (dB) levels increase as location number increases with location #1 having an average dB level of 44.8 and location #4 having an average dB level of 84.2. As participants are taken to each location in increasing order, this could have an effect on how participants anticipate noise pollution (dB) levels changing. If a participant anticipates increasing noise pollution (dB) levels, this will affect how they score on the questionnaire and how their heart reacts to the stimuli. In order to negate this, one could (a) blindfold the participants and have them wear noise-canceling headphones and (b) randomize the testing order. Since visual and auditory perception plays a big role in how some participants may react to certain situations, introducing these measures could help mitigate that.

Additionally, in a controlled environment, there would be no walking time between different testing locations. In our research design, participants were asked to walk between each of the four locations; this could have artificially skewed their questionnaire responses and raised

their Resting Heart Rate (RHR) readings. In a controlled environment, all of the data could be collected in one location. This would allow for (a) more testing to be conducted at a much wider range of noise pollution (dB) levels, (b) testing to be completed in a shorter amount of time, and (c) removal of different factors potentially affecting results. In our research, completion of all trials for each participant took ~1 hour. Without the need for walking between each location, this could be drastically reduced. As well as that, initial dB levels would not have to be collected since dB levels could be controlled. The natural variability in noise in an external environment would be negated, too. While conducting the testing in a controlled environment removes the idea of *natural* noise pollution, it could reduce the number of limitations and tighten research design.

References

- Al-Mutairi, N.Z., Al-Attar, M.A. & Al-Rukaibi, F.S. Traffic-generated noise pollution: exposure of road users and populations in Metropolitan Kuwait. *Environ Monit Assess* 183, 65–75 (2011). <https://doi.org/10.1007/s10661-011-1906-0>
- Bronzaft, A.L. and McCarthy, D. (1975) The Effect of Elevated Train Noise on Reading Ability. *Environment and Behavior*, 5, 517-528. <https://doi.org/10.1177/001391657500700406>
- Bureau of Transportation. (2018). National Transportation Noise Map. Retrieved March 23, 2022, from <https://www.bts.gov/geospatial/national-transportation-noise-map>
- CDC. (2018, February 6). *Reducing noise exposure: Guidance & Regulations*. Centers for Disease Control and Prevention. Retrieved April 12, 2022, from <https://www.cdc.gov/niosh/topics/noise/reducenoiseexposure/regsguidance.html>
- Farooqi, Z. U. R., Sabir, M., Latif, J., Aslam, Z., Ahmad, H. R., Ahmad, I., ... & Ilić, P. (2020). Assessment of noise pollution and its effects on human health in industrial hub of Pakistan. *Environmental Science and Pollution Research*, 27(3), 2819-2828.
- Geravandi, S., Takdastan, A., Zallaghi, E., Niri, M. V., Mohammadi, M. J., Saki, H., & Naiemabadi, A. (2015). Noise pollution and health effects. *Jundishapur Journal of Health Sciences*, 7(1).
- Goines, L., & Hagler, L. (2007). Noise pollution: a modern plague. *South Med J*, 100(3), 287-94. https://docs.wind-watch.org/Goines-Hagler-2007-Noise_pollution_a_modern_plague.pdf

Gupta, A., Gupta, A., Jain, K., & Gupta, S. (2018). Noise pollution and impact on children health. *The Indian Journal of Pediatrics*, 85(4), 300-306.

<https://link.springer.com/article/10.1007/s12098-017-2579-7>

Haines MM, Stansfeld SA, Job RF, Berglund B, Head J. Chronic aircraft noise exposure, stress responses, mental health and cognitive performance in school children. *Psychol Med*. 2001 Feb;31(2):265-77. doi: 10.1017/s0033291701003282. PMID: 11232914.

InformedHealth.org [Internet]. Cologne, Germany: Institute for Quality and Efficiency in Health Care (IQWiG); 2006-. Hearing loss and deafness: Normal hearing and impaired hearing. 2008 May 15 [Updated 2017 Nov 30]. Available from:

<https://www.ncbi.nlm.nih.gov/books/NBK390300/>

Morata TC, Dunn DE, Kretschmer LW, Lemasters GK, Keith RW. Effects of occupational exposure to organic solvents and noise on hearing. *Scand J Work Environ Health*. 1993 Aug;19(4):245-54. doi: 10.5271/sjweh.1477. PMID: 8235513.

https://www.sjweh.fi/show_abstract.php?abstract_id=1477

Rabinowitz, P. (2000). Noise-induced hearing loss. *American family physician*, 61(9), 2749-2756. <https://www.aafp.org/afp/2000/0501/p2749.html>

Riopel, L. (2022, March 23). *What is the Positive and Negative Affect Schedule? (PANAS)*.

PositivePsychology.com. Retrieved April 29, 2022, from

<https://positivepsychology.com/positive-and-negative-affect-schedule-panas/>

SkyPaw Co. Ltd. (2021). *Decibel X: dB Sound Level Meter* (Version 9.4.0) [Mobile App]. App Store. <https://apps.apple.com/us/app/decibel-x-db-sound-level-meter/id448155923>

Sforza, E., Chapotot, F., Lavoie, S., Roche, F., Pigeau, R., & Buguet, A. (2004). Heart rate activation during spontaneous arousals from sleep: effect of sleep deprivation. *Clinical Neurophysiology*, 115(11), 2442-2451.

<https://www.sciencedirect.com/science/article/abs/pii/S1388245704002317>

Strava, Inc. (2021). *Strava: Run, Ride, Swim* (Version 253.0.1) [Mobile App]. App Store.

<https://apps.apple.com/us/app/strava-run-ride-swim/id426826309>

APPENDIX A

Survey Questions

This scale consists of a number of words that describes different feelings and emotions. Read through each item and indicate to what extent you are experiencing the feeling or emotion at the moment.

1
very slightly
or not at all

2
a little

3
moderately

4
quite a bit

5
extremely

*

	1. very slightly or not at all	2. a little	3. moderately	4. quite a bit	5. extremely
Interested	☐	☐	☐	☐	☐
Distressed	☐	☐	☐	☐	☐
Excited	☐	☐	☐	☐	☐
Upset	☐	☐	☐	☐	☐
Strong	☐	☐	☐	☐	☐
Guilty	☐	☐	☐	☐	☐

Scared	☐	☐	☐	☐	☐
Hostile	☐	☐	☐	☐	☐
Enthusiastic	☐	☐	☐	☐	☐
Proud	☐	☐	☐	☐	☐
Irritable	☐	☐	☐	☐	☐
Alert	☐	☐	☐	☐	☐
Ashamed	☐	☐	☐	☐	☐
Inspired	☐	☐	☐	☐	☐
Nervous	☐	☐	☐	☐	☐
Determined	☐	☐	☐	☐	☐
Attentive	☐	☐	☐	☐	☐
Jittery	☐	☐	☐	☐	☐
Active	☐	☐	☐	☐	☐
Afraid	☐	☐	☐	☐	☐

APPENDIX B

Raw Participant dB and RHR Data

Participant_ID	Minimum_dB	Maximum_dB	Average_dB	RHR_min1	RHR_min2	RHR_min3	RHR_min4	RHR_min5	RHR_avg	Participant_ID
0010-1	34.8	60.6	45.2	63	67	63	63	65	64.2	0010-1
0010-2	51.9	70.0	58.7	64	63	63	70	75	67.0	0010-2
0010-3	54.6	75.3	64.1	77	74	70	71	85	75.4	0010-3
0010-4	57.7	88.7	74.2	63	63	70	69	64	65.8	0010-4
0020-1	37.3	60.9	42.6	60	59	57	57	60	58.6	0020-1
0020-2	49.0	76.0	53.9	73	70	66	67	69	69.0	0020-2
0020-3	59.9	88.8	70.9	77	72	75	71	80	75.0	0020-3
0020-4	56.1	90.5	78.8	78	76	80	78	75	77.4	0020-4
0030-1	39.2	64.3	45.8	61	62	63	61	63	62.0	0030-1
0030-2	58.4	87.8	72.0	69	73	69	67	73	70.2	0030-2
0030-3	60.6	76.7	63.9	68	67	65	68	64	66.4	0030-3
0030-4	72.8	79.6	94.2	61	62	71	62	69	65.0	0030-4
0040-1	39.2	64.3	45.8	83	75	77	83	80	79.6	0040-1
0040-2	58.4	87.8	72.0	89	79	88	87	74	83.4	0040-2
0040-3	60.6	76.7	63.9	91	101	90	99	90	94.2	0040-3
0040-4	72.8	79.6	94.2	88	77	89	87	74	83.0	0040-4
0050-1	39.5	61.1	43.2	74	78	76	74	73	75.0	0050-1
0050-2	51.2	83.1	67.1	80	78	95	76	79	81.6	0050-2
0050-3	61.9	81.1	67.3	90	78	85	81	83	83.4	0050-3
0050-4	76.2	100.7	82.3	78	80	83	80	73	78.8	0050-4

0060-1	37.8	59.2	46.8	73	75	72	78	71	73.8	0060-1
0060-2	53.5	79.8	61.2	80	76	80	73	76	77.0	0060-2
0060-3	65.3	84.9	75.8	78	68	75	79	69	73.8	0060-3
0060-4	68.2	95.3	85.3	88	86	87	81	83	85.0	0060-4
0070-1	37.8	59.2	46.8	73	61	65	67	62	65.6	0070-1
0070-2	53.5	79.8	61.2	74	62	68	69	79	70.4	0070-2
0070-3	65.3	84.9	75.8	71	68	73	75	70	71.4	0070-3
0070-4	68.2	95.3	85.3	81	80	83	95	92	86.2	0070-4
0080-1	37.8	59.7	44.2	68	68	64	67	70	67.4	0080-1
0080-2	50.4	68.2	59.8	70	71	70	68	73	70.4	0080-2
0080-3	57.3	78.9	67.2	72	70	71	70	69	70.4	0080-3
0080-4	67.8	95.4	82.4	71	74	75	78	70	73.6	0080-4
0090-1	37.8	59.7	44.2	65	64	68	61	60	63.6	0090-1
0090-2	50.4	68.2	59.8	67	68	65	64	65	65.8	0090-2
0090-3	57.3	78.9	67.2	68	70	68	71	65	68.4	0090-3
0090-4	67.8	95.4	82.4	71	75	68	67	72	70.6	0090-4
0100-1	39.5	62.0	43.4	62	67	65	62	61	63.4	0100-1
0100-2	53.6	73.4	61.7	68	67	65	69	70	67.8	0100-2
0100-3	57.3	78.9	67.2	73	75	68	66	69	70.2	0100-3
0100-4	67.8	95.4	82.4	71	75	76	78	68	73.6	0100-4

APPENDIX C

Raw Participant PANAS Data

Participant_ID	Interested	Distressed	Excited	Upset	Strong	Guilt	Scared	Hostile	Enthusiastic	Proud	Irritable	Alert	Ashamed	Inspired	Nervous	Determined	Attentive	Jittery	Active	Afraid	PAS	NAS
0010-1	4	1	2	1	4	1	1	1	3	3	1	3	1	3	1	4	4	2	2	1	32	11
0010-2	3	2	2	2	3	1	1	1	2	2	3	3	1	1	2	3	2	3	1	1	22	17
0010-3	2	2	1	2	2	2	1	2	1	2	3	3	2	1	3	2	2	3	2	1	18	21
0010-4	2	3	2	2	2	1	1	2	2	2	3	3	1	2	2	2	3	4	3	1	23	20
0020-1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	1	2	1	1	1	1	12	10
0020-2	2	1	1	1	1	1	1	1	2	1	1	2	1	2	1	1	2	1	2	1	16	10
0020-3	2	1	3	1	1	1	1	1	1	1	2	2	1	1	2	2	2	2	1	1	16	13
0020-4	1	2	1	1	1	1	2	3	2	2	2	3	1	1	2	1	2	2	2	2	16	18
0030-1	4	1	3	1	3	1	1	1	2	4	1	2	1	3	1	3	4	1	3	1	31	10
0030-2	4	2	2	1	3	1	1	1	4	4	2	4	1	2	1	2	4	2	4	1	33	13
0030-3	3	1	3	1	2	1	1	1	4	4	2	2	1	3	2	2	3	1	3	1	29	12
0030-4	4	2	3	1	3	1	2	1	3	3	3	4	1	2	2	3	4	1	3	2	32	16
0040-1	4	1	3	1	1	1	1	1	4	3		2	1	3	3	3	4	3	2	1	29	13
0040-2	3	2	3	1	3	1	1	1	4	3	1	2	1	3	2	3	3	4	3	1	30	15
0040-3	4	2	3	1	2	1	1	1	3	2	1	3	1	2	4	3	3	4	2	1	27	17
0040-4	3	2	3	1	2	1	1	1	3	2	1	3	1	3	2	2	3	3	2	1	26	14
0050-1	4	3	3	2	3	1	2	1	4	4	4	4	2	4	3	3	4	4	3	1	36	23
0050-2	4	3	3	2	2	1	1	1	3	3	1	3	1	1	2	2	3	4	3	1	27	17
0050-3	2	3	2	2	3	1	3	2	3	3	3	4	1	2	4	2	4	4	2	3	27	26
0050-4	3	3	3	2	2	1	1	1	3	3	3	4	1	2	2	2	4	4	3	1	29	19
0060-1	3	1	4	1	2	1	1	1	3	3	1	2	1	2	1	2	3	1	2	1	26	10
0060-2	3	1	3	1	2	1	1	1	4	3	1	3	1	3	1	3	3	2	2	1	29	11
0060-3	3	2	2	1	1	1	1	1	3	1	2	3	1	2	1	1	3	1	2	1	21	12
0060-4	3	1	4	1	2	1	1	1	3	1	1	2	1	3	1	3	3	1	2	1	26	10
0070-1	3	1	2	1	2	1	1	1	2	3	1	2	1	2	1	3	3	1	2	1	24	10
0070-2	2	2	2	1	4	1	1	3	3	3	1	4	1	4	1	4	4	1	3	1	33	13
0070-3	2	1	1	1	3	1	1	2	2	3	4	4	1	1	2	1	3	1	3	1	23	15
0070-4	2	1	3	1	3	1	1	2	4	4	1	4	1	3	1	3	4	1	4	1	34	11
0080-1	3	1	2	1	1	1	1	1	3	3	1	3	1	2	2	3	4	2	2	1	26	12
0080-2	2	2	3	1	2	1	1	1	3	1	1	2	1	2	2	3	3	2	3	1	24	13
0080-3	2	3	2	2	2	2	1	2	1	1	2	3	1	1	3	2	2	3	3	1	19	20
0080-4	3	2	3	2	2	1	2	2	2	1	3	2	1	1	2	1	2	3	2	1	19	19
0090-1	3	1	3	1	2	1	1	2	4	4	2	3	1	2	2	4	3	1	2	1	30	13
0090-2	2	1	3	1	1	1	1	1	2	2	1	2	1	3	2	2	3	2	3	1	23	12
0090-3	2	3	2	1	1	1	2	3	2	2	3	2	1	1	2	2	3	2	2	2	19	20
0090-4	2	2	2	1	2	1	2	3	2	2	3	2	1	1	2	3	3	2	2	2	21	19
0100-1	3	1	3	1	2	1	1	1	3	2	2	3	1	1	1	2	4	2	2	1	25	12
0100-2	3	2	3	1	2	1	1	1	2	1	1	2	1	3	2	3	3	1	1	1	23	12
0100-3	2	2	2	1	2	1	1	2	2	3	2	3	1	1	3	2	3	3	2	3	22	19
0100-4	1	1	2	1	3	1	1	2	2	1	3	3	1	1	2	2	2	3	3	2	20	17